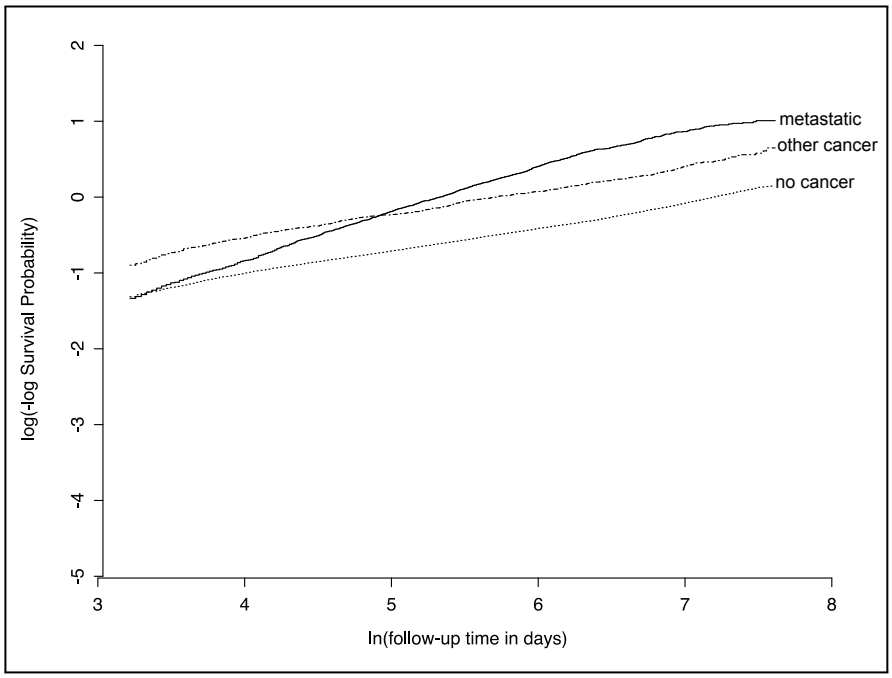


Figure 5. Example of violation of proportional hazards assumption in SUPPORT study dataset. For the "ca" variable, patients with metastatic cancer have a similar risk of early death as other patients, but a higher risk of late death, as evidenced by non-parallel lines on this plot of log(-log survival) over time.

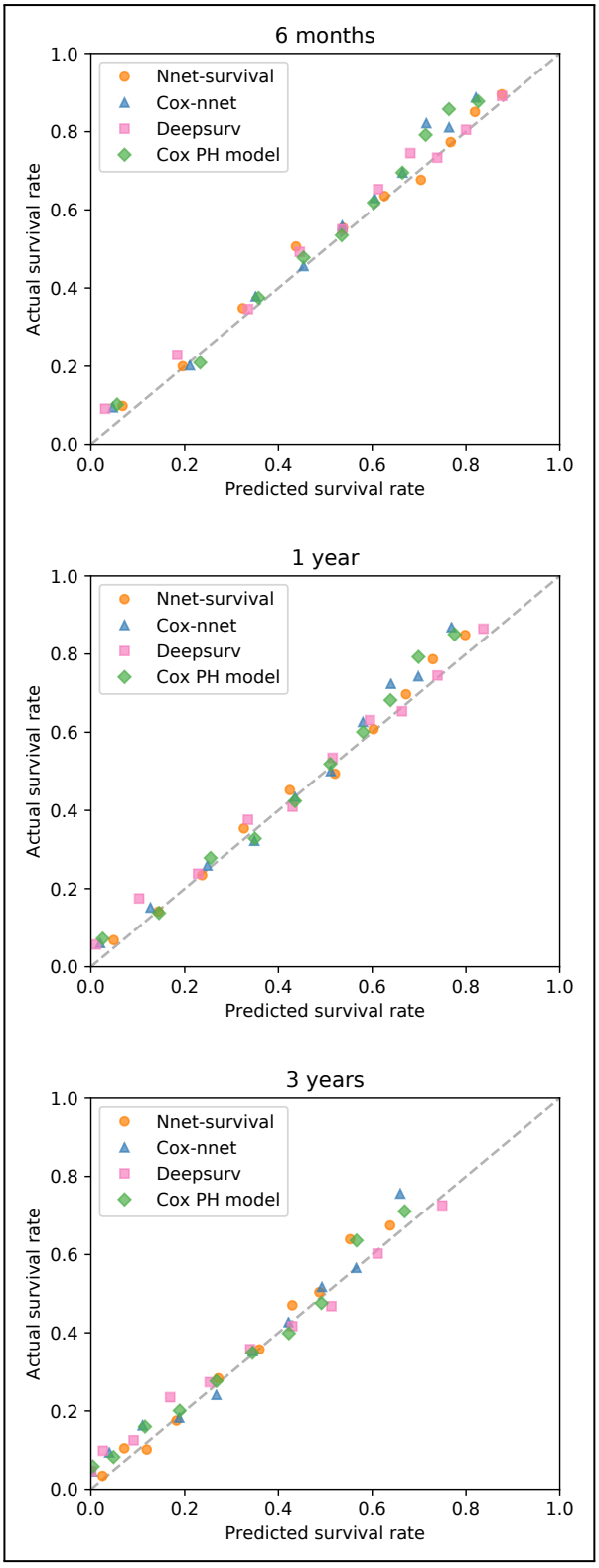


Figure 6. Calibration of four survival models on SUPPORT study test set. For each model, for each of three follow-up times, patients were grouped by deciles of predicted survival probability. Then, for each decile, mean actual survival (Kaplan-Meier method) was plotted against mean predicted survival. A perfectly calibrated model would have all points on the identity line (dashed).

We compared running time of the three neural network models for various training set sizes, with results shown in Fig. 7. Simulated datasets of size 1,000 to 1,000,000 were created by sampling from the SUPPORT study dataset with replacement. Each model was run for 1,000 epochs. For sample sizes of 100,000 and higher, the Cox-nnet model ran out of memory on a computer with 32 GB memory; therefore, for this model running times could only be calculated for sample sizes of 1,000 to 31,622.

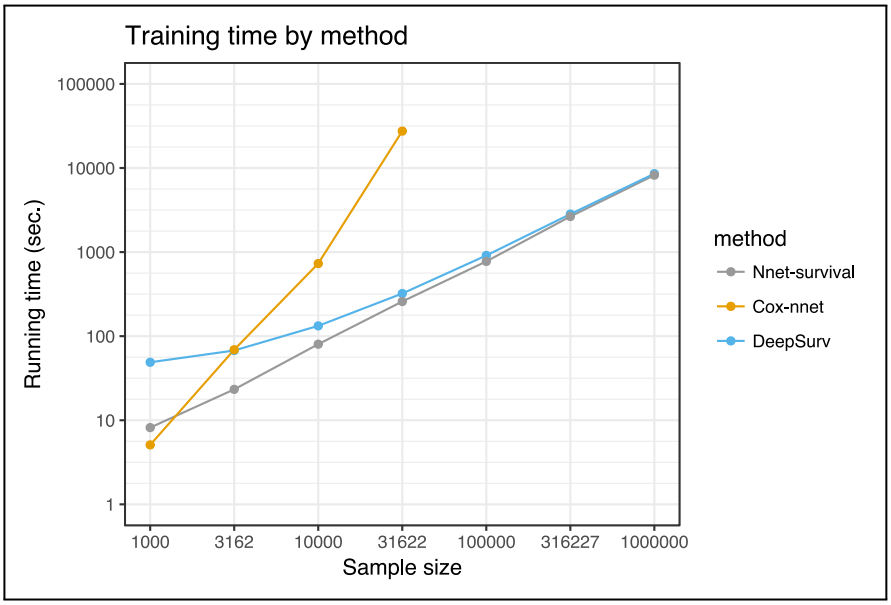


Figure 7. Running time of the three neural network models on SUPPORT study dataset. Each point represents the average of three runs. Cox-nnet ran out of memory for sample sizes of 100,000 and higher.

DISCUSSION

We presented Nnet-survival, a discrete-time survival model for neural networks. It is theoretically justified since the likelihood function is used as the loss function, and naturally incorporates non-proportional hazards. Because it is a parametric model, it can be trained with minibatch gradient descent as the likelihood/loss depends only on the patients in the current minibatch. This enables fast training, use on datasets that do not fit in memory, and can avoid the network getting stuck in a local minimum of the loss function (Bottou, 1991). This is in contrast to models based on the Cox proportional hazards model such as Cox-nnet (Ching et al., 2018) and Deepsurv (Katzman et al., 2018), which require the entire training set to be used for each model update (batch gradient descent). The Nnet-survival model can be applied to a variety of neural network architectures, including multilayer perceptrons and convolutional neural networks.

In our experiments, the model performed well on both simulated and real datasets. It was challenging to find a publicly available dataset that would potentially highlight the advantages of the model. Ideally, such a dataset would have large sample size, predictor data such as images or text that are well-suited to neural networks, and time-to-event outcome data. Since no such dataset was available to our knowledge, we used the SUPPORT study dataset of 9,105 hospitalized patients, which has moderate sample size and time-to-event outcome data, but has low-dimensional predictor data that may not result in a benefit from a neural network approach. For this dataset, our model's discrimination and calibration performance was similar to several other neural network survival models and a traditional Cox proportional hazards model. In

running time tests, its running time was similar to Deepsurv (Katzman et al., 2018) and better than Cox-nnet (Ching et al., 2018) for sample sizes >1000. Interestingly, Cox-nnet ran out of memory for larger dataset sizes, because it stores an n by n matrix where n is the sample size (variable name *R_matrix_train* in the Cox-nnet code).

While our model has several advantages and we think it will be useful for a broad range of applications, it does has some drawbacks. The discretization of follow-up time results in a less smooth predicted survival curve compared to a non-discrete parametric survival model such as a Weibull accelerated failure time model. As long as a sufficient number of time intervals is used, this is not a large practical concern — for instance, with 19 intervals the curves in Fig. 6 appear very smooth. Unlike a parametric survival model, the model does not provide survival predictions past the end of the last time interval, so it is recommended to extend the last interval past the last follow-up time of interest.

The advantages of parametric survival models and our discrete-time survival model could be combined in the future using a flexible parametric model, such as the cubic spline-based model of Royston and Parmar, implemented in the *flexsurv* R package(Royston and Parmar, 2002; Jackson, 2016). Complex non-proportional hazards models can be created in this way, and likely could be implemented in deep learning packages.

CONCLUSIONS

Our discrete-time survival model allows for non-proportional hazards, can be used with stochastic gradient descent, allows rapid training time, and was found to produce good discrimination and calibration performance with both simulated and real data. For these reasons, it may be useful to medical researchers.

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